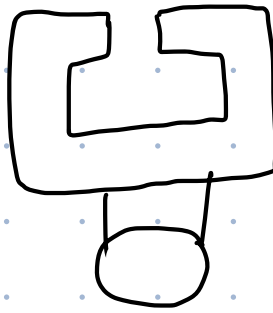
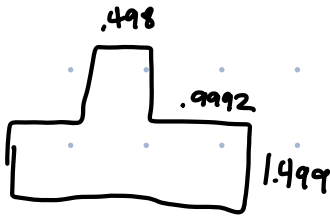


$$F.S = 4$$

$$F = 650 \text{ lbf}$$

Ti-6Al-V4



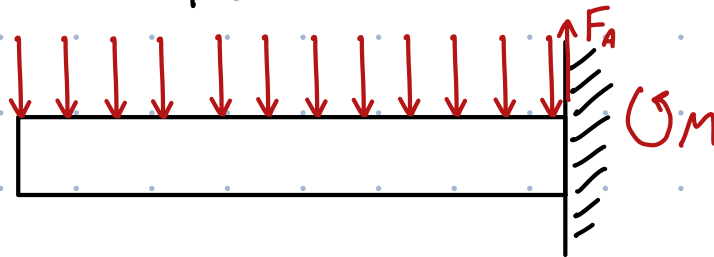
A) Knowns: $F = 650 \text{ lbf}$ $F.S = 4$ Mat. Ti-6Al-V4

$$E = 114 \text{ GPa} = 16534.3 \text{ ksi} \quad S_y = 828 \text{ MPa} = 120.091 \text{ ksi}, \text{ Max deflection } 0.05 \text{ in}$$

Assumptions: $L = 1 \text{ in}$, No failure from Shear, Treated as a Cantilever Beam with a distributed load.

Unknowns: Cross Sectional Area, Section Moduli

FBP:

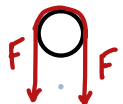


Stress Analysis

$$Z = S.F. \cdot W \cdot \frac{L}{2S_y} \quad r = \left[4 \cdot \left(\frac{Z}{\pi} \right) \right]^{1/3}$$

$$Z = (4) \cdot (1300) \cdot \frac{1 \text{ in}}{2(120091)} = .02165 \text{ in}$$

note: $W = 2F$



$$4 \cdot \left(\frac{.02165}{\pi} \right)$$

$$r = .302$$

$$d = .604 \text{ in}$$

Stiffness Analysis

$$\delta = \frac{WL^3}{8EI}$$

$$Z = .02165 \text{ in}$$

$$I = \frac{\pi r^4}{4}$$

$$\delta = \frac{WL^3}{8E \left(\frac{\pi r^4}{4} \right)}$$

$$r = \left[\frac{WL^3}{82E\pi} \right]^{1/4} = \left[\frac{(1300) (1)^3 (4)}{(.005)(2)(16534.3 \times 10^3)(\pi)} \right]^{1/4}$$

$$r = .316 \text{ in}$$

$$d = .632 \text{ in}$$

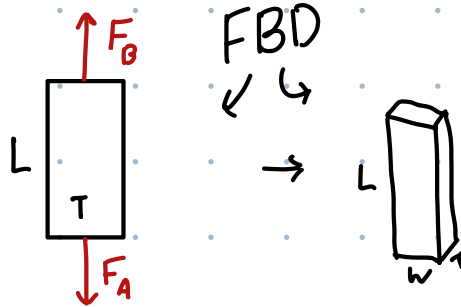
B) Knowns: $F = 650 \text{ lbf}$ $F.S = 4$ Mat. Ti-6Al-V4

$E = 114 \text{ GPa} = 16534.3 \text{ ksi}$ $S_y = 828 \text{ MPa} = 120.091 \text{ ksi}$, Max deflection $\leq .005 \text{ in}$

Assumptions: $L = .5 \text{ in}$, $W = .604 \text{ in OR } .632 \text{ in}$, Axial Load

Unknowns: Thickness

Based on Stress
vs
Stiffness



Stress Analysis

$$\sigma = \frac{P}{A} \quad A = W \cdot T \quad P = F \cdot 2$$

$$\Rightarrow \frac{2F}{W \cdot T} \xrightarrow{F.S} \frac{2F(F.S)}{W \cdot T} = S_y$$

$$\Rightarrow T = \frac{2F(F.S)}{W \cdot S_y}$$

$$T = \frac{2(650)(4)}{(.604 \text{ in})(120091 \text{ lb/in}^2)} = .0717 \text{ in Thick}$$

Stiffness Analysis

$$\delta = \frac{PL}{AE} \quad A = T \cdot W \quad P = F \cdot 2 \cdot F.S$$

$$\hookrightarrow \delta = \frac{2FLF.S}{T \cdot W \cdot E}$$

$$T = \frac{2F(F.S)L}{E \cdot W \cdot \delta}$$

$$\Rightarrow \frac{2(650)(4)(.5)}{(16534.3 \times 10^3)(.632)(.005)}$$

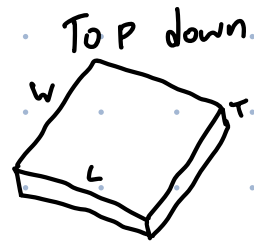
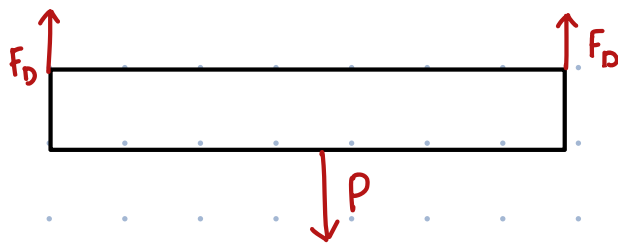
$$.0498 \text{ in Thick}$$

C) Knowns: $F = 650 \text{ lbf}$ $F.S = 4$ Mat. Ti-6Al-V4

$E = 114 \text{ GPa} = 16534.3 \text{ ksi}$ $S_y = 828 \text{ MPa} = 120.091 \text{ ksi}$, Max deflection $\leq .005 \text{ in}$

Assumptions: $L = 2.5 \text{ in}$, $W = 1 \text{ in}$, Simply Supported Be

Unknowns: Thickness



Stress Analysis

$$\sum F_y = 2F_D - P = 0 \therefore \frac{1}{2}P = F_D$$

$$P = 2 \cdot F = 1300 \quad F_D = 650 \text{ lbf}$$

$$\sigma = \frac{My}{I}, \quad I = \frac{b \cdot h^3}{12} \quad h = T$$

$$M = r \times F, \quad r = \frac{1}{2}L, \quad F = P \rightarrow M = \frac{1}{4}L \cdot P$$

$$\sigma = \frac{\frac{1}{4}L \cdot P \cdot F.S. \cdot \frac{1}{2}h}{\frac{b \cdot h^3}{12}} \rightarrow \sigma = \frac{15 \cdot L \cdot P \cdot F.S.}{b \cdot h^2}$$

$$\Rightarrow h = \left[\frac{15 \cdot L \cdot P \cdot F.S.}{b \cdot \sigma} \right]^{1/2}$$

$$\Rightarrow \left[\frac{(1.5)(2.5)(1300)(4)}{(1)(120001)} \right]^{1/2} T = .403 \text{ in} \\ .403 \text{ in Thick}$$

Stiffness Analysis

$$\delta = \frac{Pl^3}{48EI} \quad I = \frac{W h^3}{12}$$

$$\frac{Pl^3}{4EIh^3} = \delta, \quad h = \left[\frac{Pl^3}{4E \cdot W \cdot \delta} \right]^{1/3}$$

$$h = \left[\frac{(1300)(4)(2.5)}{(4)(16534.3 \times 10^3)(1)(.005)} \right]^{1/3}$$

$$h = .340 \text{ in}$$

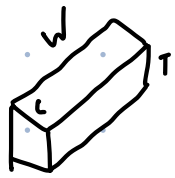
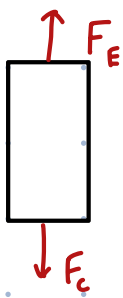
$$.340 \text{ in Thick}$$

D) Knowns: $F = 650 \text{ lbf}$ $F.S. = 4$ Mat. Ti-6Al-V4

$$F_c = 650 \text{ lbf} \quad E = 114 \text{ GPa} = 16534.3 \text{ ksi} \quad S_y = 828 \text{ MPa} = 120.091 \text{ ksi} \quad \text{Max deflection } \sigma = .005 \text{ in}$$

Assumptions: $W = 1 \text{ in}$ $T = 1.5 \text{ in}$, Axial Loaded

Unknowns: Length



Stress Analysis

$$\sigma = \frac{F}{A} \quad A = W \cdot L \quad F = F_c = F_E$$

$$S_y = \frac{F_c}{W \cdot L} \rightarrow L = \frac{F_c \cdot F.S}{W \cdot S_y}$$

$$L = \frac{650(4)}{(11)(120091)} = .0217 \text{ in}$$

Stiffness Analysis

$$\delta = \frac{PT}{AE} \quad P = F_c = F_E$$

$$A = W \cdot L$$

$$\delta = \frac{F_c \cdot T \cdot F.S}{W \cdot L \cdot E} \quad L = \frac{(650)(4)(1.5)}{(11)(.005)(16534.3 \times 10^3)}$$

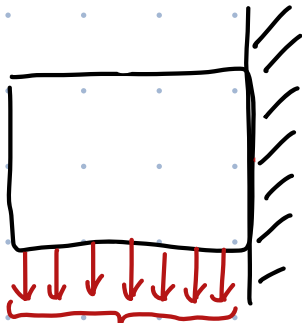
$$L = \frac{F_c \cdot T \cdot F.S}{W \cdot E \cdot \delta} \quad L = .0471 \text{ in}$$

E) Knowns: $F = 650 \text{ lbf}$ $F.S = 4$ Mat. Ti-6Al-V4

$F_c = 650 \text{ lbf}$ $E = 114 \text{ GPa} = 16534.3 \text{ ksi}$ $S_y = 828 \text{ MPa} = 120.091 \text{ ksi}$ Max deflection 0.005 in

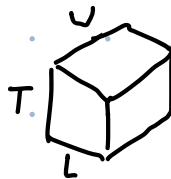
Assumptions: $W = 1 \text{ in}$, $L = 1 \text{ in}$, Cantilever Beam

Unknowns: Thickness



$P/2$

↑ note: P is split among
two blocks $\therefore P/2 = F$



Stress Analysis

Stiffness Analysis

$$\sigma = \frac{M_y}{I}$$

$$M = P/2 \times L/2$$

$$Y = T/2$$

$$I = \frac{W T^3}{12}$$

$$S_y = \frac{P \cdot L \cdot T}{8 \frac{W T^3}{12}} \rightarrow \frac{(1.5)(P)(L)(fs)}{W T^2}$$

$$T = \left[\frac{(1.5)(1300)(4)(1)}{(1)(120091)} \right]^{1/2} = .255 \text{ in Thick}$$

$$\delta = \frac{W L^4}{8 E I}$$

$$W = 650 \text{ lb.in}$$

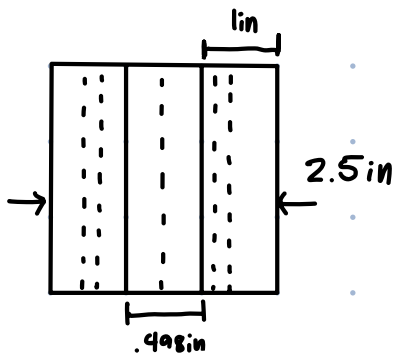
$$I = \frac{W T^3}{12}$$

$$\delta = \frac{W L}{8 E \frac{W T^3}{12}}$$

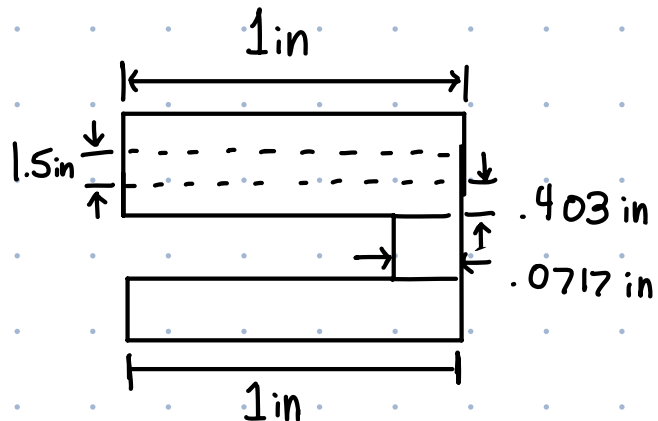
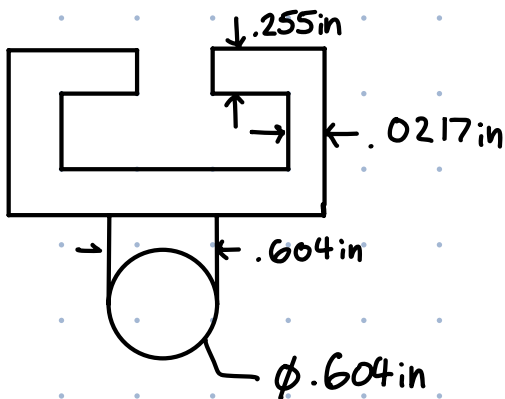
$$T = \left[\frac{1.5 W L^3 f.s}{E W 8} \right]^{1/3}$$

$$= \left[\frac{(1.5)(650)(1)^3(4)}{(16534.3 \times 10^3)(1)(.005)} \right]^{1/3}$$

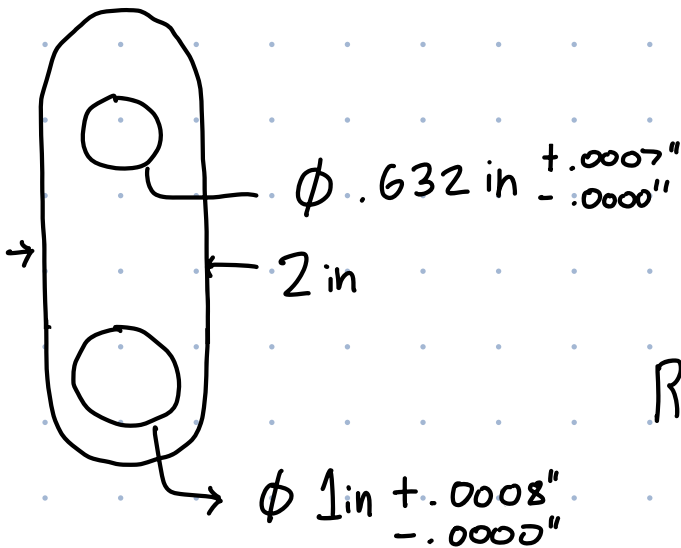
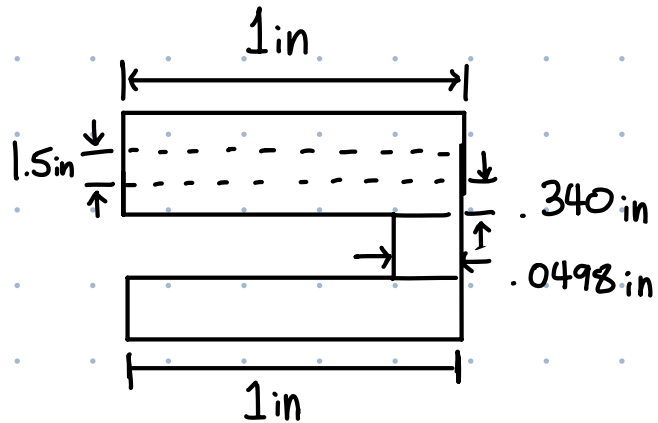
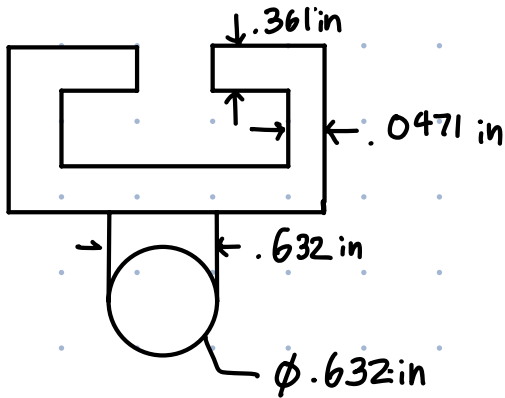
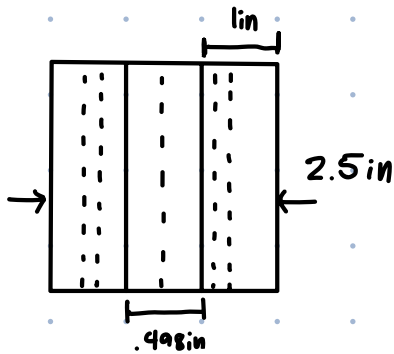
$$.361 \text{ in Thick}$$



Stress Multiview



Stiffness Multiview



1 inch @ Thinnest

RC3

Knowns: $F = 650\text{ lbf}$ $F.S = 4$ Mat. Ti-6Al-V4

$E = 114\text{ GPa} = 16534.3\text{ ksi}$

$S_y = 828\text{ MPa} = 120.091\text{ ksi}$ Max deflection $\leq .005\text{ in}$

Assumptions: 4 in long, 1 in thick, Axial Loaded

Unknowns: Thickness

Stress

$$\sigma = \frac{F}{A}$$

$$A = W \cdot T$$

$$\sigma = \frac{F}{W \cdot T} \rightarrow T = \frac{F(F.S)}{W \cdot S_y}$$

$$T = \frac{650(4)}{1(120091)}$$

.0216 in
Thick

Stiffness

$$\delta = \frac{PL}{AE}$$

$$P = F \quad A = W \cdot T$$

$$\delta = \frac{FL \cdot F.S}{W \cdot T \cdot E}$$

$$\rightarrow T = \frac{F \cdot L \cdot F.S}{W \cdot E \cdot \delta}$$

$$\Rightarrow \frac{(650)(4)(4)}{(1)(16534.3 \times 10^3)(.005)}$$

.126 in Thick